

SF-EDGE Validation Strategy

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Owner: QA and Validation

Project: Smart Factory Edge Gateway

1. Validation Objective

The objective of validation is to provide confidence that SF-EDGE can collect, buffer, forward, secure, and audit industrial telemetry under realistic operational conditions.

Validation shall focus on requirements that affect production continuity, traceability, configuration correctness, and cybersecurity.

2. Test Levels

2.1 Unit Tests

Unit tests cover:

- protocol configuration validators;
- telemetry normalization;
- retry delay calculation;
- event ID generation;
- redaction utilities;
- role permission checks.

2.2 Integration Tests

Integration tests cover:

- adapter to normalizer flow;
- normalizer to local buffer flow;
- local buffer to cloud forwarder flow;
- API to audit service flow;
- configuration versioning and rollback.

2.3 System Tests

System tests cover the gateway as a complete system using simulated devices and a simulated cloud endpoint.

2.4 Security Tests

Security tests cover authentication, authorization, TLS behavior, support grants, and diagnostic bundle redaction.

2.5 Long-Run Tests

Long-run tests cover continuous operation for at least 30 days under nominal load or a compressed endurance profile approved by QA.

3. Test Environment

The primary validation environment shall include:

- one reference gateway hardware unit;
- one simulated Modbus TCP device;
- one simulated OPC UA server;
- one MQTT broker;

- one HTTP sensor simulator;
- one cloud ingestion simulator;
- one NTP server or NTP simulator;
- log collection and metrics capture.

4. Core Validation Scenarios

VAL-SCEN-001 Device Registration

Verify that a maintenance engineer can register a valid Modbus TCP device and that duplicate device IDs are rejected.

Requirements covered:

- REQ-FUNC-001;
- REQ-FUNC-002;
- SEC-REQ-010;
- SEC-REQ-031.

VAL-SCEN-002 Telemetry Collection and Normalization

Verify that telemetry collected from Modbus TCP and OPC UA devices is normalized into the common schema.

Requirements covered:

- REQ-FUNC-003;
- REQ-NFR-005.

VAL-SCEN-003 Cloud Outage and Local Buffering

Simulate a cloud outage lasting 2 hours while devices continue sending telemetry.

Expected result:

- telemetry collection continues;
- records are buffered locally;
- no records are lost;
- forwarding resumes when the cloud simulator becomes available.

Requirements covered:

- REQ-FUNC-004;
- REQ-FUNC-005;
- REQ-NFR-003.

VAL-SCEN-004 Duplicate Prevention

Replay buffered records after a transient cloud failure and verify that duplicate telemetry records are not accepted by the cloud simulator.

Requirements covered:

- REQ-FUNC-005;
- REQ-NFR-005.

VAL-SCEN-005 Device Health Degradation

Disconnect a registered device and verify that the health state becomes disconnected within 30 seconds after failure detection.

Requirements covered:

- REQ-FUNC-006;

- REQ-FUNC-007.

VAL-SCEN-006 Configuration Versioning and Rollback

Apply a valid configuration update, then apply an invalid update, then roll back to the previous valid version.

Expected result:

- each update creates a configuration version;
- rollback restores the previous valid configuration;
- audit events remain immutable.

Requirements covered:

- REQ-FUNC-008;
- REQ-FUNC-009;
- SEC-REQ-031.

VAL-SCEN-007 Diagnostic Redaction

Generate a diagnostic bundle containing known secret-like values.

Expected result:

- secrets are redacted;
- diagnostic file does not contain private keys, passwords, or API tokens;
- export operation is audited.

Requirements covered:

- REQ-FUNC-010;
- SEC-REQ-021;
- SEC-REQ-052.

VAL-SCEN-008 Role-Based Access

Verify that:

- plant_operator cannot update device configuration;
- maintenance_engineer cannot manage users;
- site_admin can perform rollback;
- cloud_support_engineer cannot access diagnostics without a support grant.

Requirements covered:

- SEC-REQ-010;
- SEC-REQ-011.

VAL-SCEN-009 Startup Readiness

Power-cycle the gateway and verify that telemetry collection starts within 90 seconds.

Requirements covered:

- REQ-NFR-001.

VAL-SCEN-010 Buffer Capacity Alert

Fill the local buffer above 80 percent capacity and verify that a buffer capacity alert is generated.

Requirements covered:

- REQ-FUNC-004;
- REQ-FUNC-007.

5. Entry Criteria

Validation may start when:

- requirements are approved;
- API contracts are stable;
- simulated devices are available;
- test accounts are configured;
- logging and metrics capture are enabled.

6. Exit Criteria

Validation is complete when:

- all high-priority scenarios pass;
- no open critical or high defects remain;
- traceability matrix is updated;
- diagnostic bundle redaction evidence is archived;
- QA lead approves the validation report.

7. Defect Severity

Critical:

- telemetry loss;
- unauthorized administrative access;
- secrets exposed in diagnostics;
- gateway cannot start.

High:

- health state not updated;
- rollback failure;
- duplicate telemetry forwarding;
- audit event missing for privileged action.

Medium:

- incorrect alert message;
- delay above expected threshold but without data loss;
- non-critical UI inconsistency.

Low:

- cosmetic issue;
- spelling issue;
- non-blocking log formatting issue.

8. Traceability Expectation

Every validation scenario shall reference one or more requirements. Every high-priority requirement shall be covered by at least one validation scenario.

Coverage shall be reviewed before release.